

CYBERSECURITY REPORT – LABORATORY 10

1 Problems and questions - OSINT:

1.1 What is the technical idea behind OS fingerprinting?

OS fingerprinting **identifies the operating system** by observing how its **TCP/IP stack implements protocol details** as IP/TCP header fields (TTL, window size, DF flag), TCP options and order, ICMP responses and subtle quirks to crafted probes.

Those observable patterns form signatures that map to particular OSes/versions.

1.2 Why OS fingerprinting can be important for security?

Knowing the OS helps an attacker pick applicable exploits; defenders use it for asset inventory, vulnerability prioritization and detecting anomalous or unauthorized devices.

In short: it *narrows the attacker's choices and helps defenders focus patching and monitoring*.

1.3 What is a difference between passive and active OS fingerprinting?

Active fingerprinting **sends probes** to the host and analyzes the replies (accurate but noisy/easily detected).

Passive fingerprinting **only observes existing traffic** from the host (stealthy but **needs traffic** and can be less complete).

1.4 Is it possible to protect your systems from OS fingerprinting?

Partially. You can harden and **obfuscate** (packet filtering, drop/normalize suspicious probes, disable TCP timestamps, tweak banners, use NAT/IDS/IPS or stack-randomizing tools) **to make fingerprinting harder** or produce ambiguous signatures, but you generally **cannot make a host perfectly invisible to determined analysis**.

1.5 Is it possible to fool intruder and to show him that your host is not alive?

Yes, if the host simply drops all probe packets (firewall rules that silently drop ICMP/TCP probes) it will often appear “down” or “filtered.” Note a sophisticated attacker may use alternate discovery methods or correlate other signals, so this is a mitigation, not a guaranteed deception.

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1.6 What is DNS zone transfer and what is a risk related to this mechanism?

A DNS zone transfer (AXFR/IXFR) copies all zone records from a DNS master to a slave. If allowed to arbitrary outsiders, it leaks the domain's entire DNS map (hostnames, subdomains, IPs, MX records, etc.), giving attackers a ready blueprint of the network.

1.7 Is it legal to use OSINT methods to get sensitive information?

Generally, OSINT (collecting *publicly available* information) is **legal**, but legality depends on jurisdiction and context.

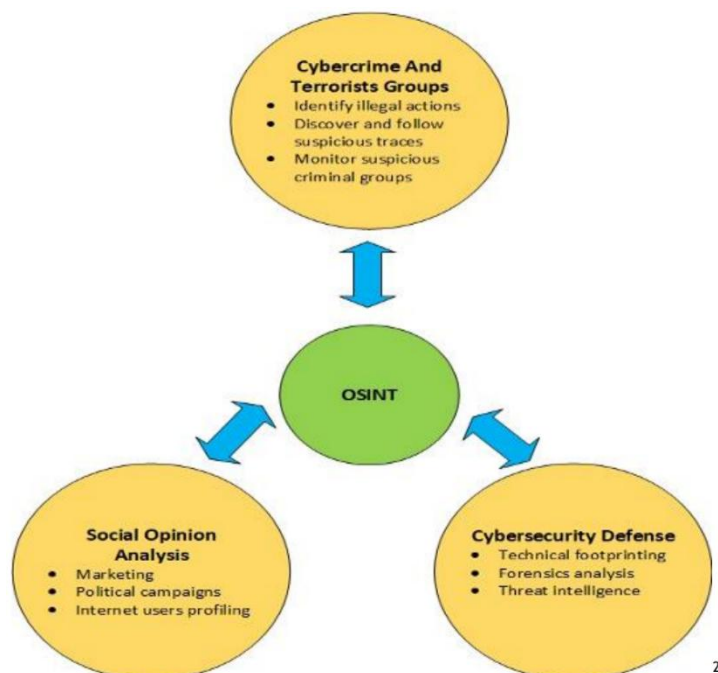
Crossing into **unauthorized access**, scraping data **against terms of service**, **privacy-law violations**, or using deceptive/illicit means is illegal. Always check laws and policies.

1.8 What is the biggest threat in context of security and OSINT methods?

Human-targeted attacks (social engineering / spear-phishing) enabled by aggregated OSINT are the largest practical threat; small public facts combined let attackers craft convincing scams that bypass technical controls.

1.9 How to protect your sensitive data from OSINT search?

Reduce public exposure: lock down privacy settings, remove unnecessary PII from web profiles and documents, harden DNS (no open zone transfers), disable verbose service banners, use **strong unique passwords** + MFA, train staff on data hygiene, monitor for leaked data and use DLP/brand-monitoring services.



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2 Tasks:

2.1 Select one well know domain (e.g. www.pwr.edu.pl) Try to gather some more specific information about the domain and its owner: e.g. who have registered the domain and when, till when it is valid, is it using cloudflare or other DDOS protection:

What to search for:

- WHOIS: registrant, registrar, creation & expiry dates, contact (GDPR may hide this).
- Name servers: indicate delegated DNS provider (Cloudflare etc).
- A/AAAA/MX/CNAME/TXT/SOA: infrastructure and mail handling.
- IP addresses and ASN: hosting provider and geographic hints.
- CDN/DDoS provider indicators: NS names, reverse DNS, certificate issuer, Cloudflare/Akamai signatures.
- Certificate transparency logs / crt.sh results for subdomains.
- Technology fingerprint (server, frameworks).

2.2 Query the whois database about that domain:

whois <domain> | tee <FileToStore>.txt

Check DNS servers as “nameservers”, its organization type, its creation date and more information of the servers.

```
(stud@kali-vm)-[~]
$ whois pwr.edu.pl | tee whois_pwr.txt
DOMAIN NAME:                pwr.edu.pl
registrant type:            organization
nameservers:                dns.pwr.edu.pl. [156.17.18.10]
                             dns2.pwr.edu.pl. [156.17.18.11]
                             dns3.pwr.edu.pl. [156.17.197.98]
                             ns1.net.icm.edu.pl.
                             ns2.net.icm.edu.pl.
                             sun2.pwr.wroc.pl. [156.17.5.2]
created:                    2004.12.17 06:30:20
last modified:              2025.12.09 15:54:05
renewal date:               2027.12.16 13:00:00

option:                     no
dnssec:                     Unsigned

REGISTRAR:
cyber_Folks S.A.
ul. Wierzbicice 1B
61-569 Poznań
Polska/Poland
Tel: +48.122963663
https://cyberfolks.pl/
domeny@cyberfolks.pl

WHOIS database responses:    https://dns.pl/en/whois

WHOIS displays data with a delay not exceeding 15 minutes in relation to the .pl Registry system
```

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2.3 Collect information about the DNS servers and the corresponding records of a target domain:

- A. Use the host command line tool to lookup the IP address of a host from a DNS server:

host www.<domain> | tee <FileToStore>.txt

```
(stud@kali-vm)-[~]
$ host www.pwr.edu.pl | tee host_www_pwr.txt
www.pwr.edu.pl has address 156.17.75.200
```

There you can see the Public Web Server hosting the university website: 153.17.75.200

host -l <domain> <nameserver>

```
(stud@kali-vm)-[~]
$ host -l pwr.edu.pl dns.pwr.edu.pl

Using domain server:
Name: dns.pwr.edu.pl
Address: 156.17.18.10#53
Aliases:

Host pwr.edu.pl not found: 5(REFUSED)
; Transfer failed.

(stud@kali-vm)-[~]
$ host -l pwr.edu.pl dns2.pwr.edu.pl

Using domain server:
Name: dns2.pwr.edu.pl
Address: 156.17.18.11#53
Aliases:

Host pwr.edu.pl not found: 5(REFUSED)
; Transfer failed.

(stud@kali-vm)-[~]
$ host -l pwr.edu.pl dns3.pwr.edu.pl

Using domain server:
Name: dns3.pwr.edu.pl
Address: 156.17.197.98#53
Aliases:

Host pwr.edu.pl not found: 5(REFUSED)
; Transfer failed.
```

```
(stud@kali-vm)-[~]
$ host -l pwr.edu.pl ns1.net.icm.edu.pl

Using domain server:
Name: ns1.net.icm.edu.pl
Address: 213.135.57.10#53
Aliases: shop

Host pwr.edu.pl not found: 5(REFUSED)
; Transfer failed.

(stud@kali-vm)-[~]
$ host -l pwr.edu.pl ns2.net.icm.edu.pl

Using domain server:
Name: ns2.net.icm.edu.pl
Address: 193.219.28.51#53
Aliases:

Host pwr.edu.pl not found: 5(REFUSED)
; Transfer failed.

(stud@kali-vm)-[~]
$ host -l pwr.edu.pl sun2.pwr.wroc.pl

Using domain server:
Name: sun2.pwr.wroc.pl
Address: 156.17.5.2#53
Aliases:

Host pwr.edu.pl not found: 5(REFUSED)
; Transfer failed.
```

For all 6 authoritative nameservers: They refused the AXFR zone transfer and responded with error code **5 (REFUSED)**. This is the **correct and secure behavior**. As they do **not expose internal DNS records**.

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B. Use the dig command to do DNS interrogation

dig example.com any

dig @8.8.8.8 example.com

dig @8.8.8.8 example.com MX

dig -x 8.8.8.8 dig example.com +trace

dig +noall +answer

These dig queries are **passive OSINT methods** to collect **publicly visible technical info** about a domain: IPs, mail servers, authoritative name servers, subdomains, and hostnames. This info is used in **information gathering / target reconnaissance** before penetration testing or social engineering.

Run: > dig pwr.edu.pl any | tee dig_any.txt

```
(stud@kali-vm)-[~]
$ dig pwr.edu.pl any | tee dig_any.txt

; <<>> DiG 9.19.21-1+b1-Debian <<>> pwr.edu.pl any
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 19423
;; flags: qr rd ra; QUERY: 1, ANSWER: 14, AUTHORITY: 0, ADDITIONAL: 11

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: 44da35372cbb854c6ce0cbc8693c68384b6f4326c60b42e4 (good)
;; QUESTION SECTION:
;pwr.edu.pl.                IN      ANY

;; ANSWER SECTION:
pwr.edu.pl.                5568    IN      MX      5 alt1.aspmx.l.google.com.
pwr.edu.pl.                5568    IN      MX      15 alt4.aspmx.l.google.com.
pwr.edu.pl.                5568    IN      MX      7 alt2.aspmx.l.google.com.
pwr.edu.pl.                5568    IN      MX      10 alt3.aspmx.l.google.com.
pwr.edu.pl.                5568    IN      MX      0 aspmx.l.google.com.
pwr.edu.pl.                5568    IN      A       156.17.75.200
pwr.edu.pl.                80525   IN      NS      dns4.pwr.edu.pl.
pwr.edu.pl.                80525   IN      NS      dns3.pwr.edu.pl.
pwr.edu.pl.                80525   IN      NS      dns.pwr.wroc.pl.
pwr.edu.pl.                80525   IN      NS      sun2.pwr.wroc.pl.
pwr.edu.pl.                80525   IN      NS      dns2.pwr.edu.pl.
pwr.edu.pl.                80525   IN      NS      ns1.net.icm.edu.pl.
pwr.edu.pl.                80525   IN      NS      dns.pwr.edu.pl.
pwr.edu.pl.                80525   IN      NS      ns2.net.icm.edu.pl.

;; ADDITIONAL SECTION:
ns1.net.icm.edu.pl.        58171   IN      A       213.135.57.10
dns.pwr.wroc.pl.          50966   IN      A       156.17.18.10
dns3.pwr.edu.pl.          1325    IN      A       156.17.197.98
ns2.net.icm.edu.pl.        72683   IN      A       193.219.28.51
dns.pwr.edu.pl.           1325    IN      A       156.17.18.10
dns4.pwr.edu.pl.          1325    IN      A       156.17.140.4
sun2.pwr.wroc.pl.         1034    IN      A       156.17.5.2
dns2.pwr.edu.pl.          1325    IN      A       156.17.18.11
ns1.net.icm.edu.pl.        58171   IN      AAAA    2001:6a0:0:2000::10
ns2.net.icm.edu.pl.        72683   IN      AAAA    2001:6a0:0:2002::51

;; Query time: 4 msec
;; SERVER: 10.138.0.1#53(10.138.0.1) (TCP)
;; WHEN: Fri Dec 12 20:08:39 CET 2025
;; MSG SIZE rcvd: 550
```

Shows all **publicly available DNS records** (MX, NS, A) in one query. OSINT use: identify **mail servers**, **name servers**, and **IP addresses** that could be targeted for further info or mapping the network.

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Run: > dig @8.8.8.8 pwr.edu.pl | tee dig_google_pwr.txt

```
(stud@kali-vm)-[~]
$ dig @8.8.8.8 pwr.edu.pl | tee dig_google_pwr.txt

; <<>> DiG 9.19.21-1+b1-Debian <<>> @8.8.8.8 pwr.edu.pl
; (1 server found)
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 21845
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;pwr.edu.pl.                IN      A

;; ANSWER SECTION:
pwr.edu.pl.                3397    IN      A      156.17.75.200

;; Query time: 27 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Fri Dec 12 20:18:48 CET 2025
;; MSG SIZE rcvd: 55
```

Confirms the **current public A record** via a recursive resolver. OSINT use: check **publicly visible IP**, see what external users see, detect caching or CDN.

Run: > dig @8.8.8.8 pwr.edu.pl MX | tee dig_google_MX_pwr.txt

```
(stud@kali-vm)-[~]
$ dig @8.8.8.8 pwr.edu.pl MX | tee dig_google_MX_pwr.txt

; <<>> DiG 9.19.21-1+b1-Debian <<>> @8.8.8.8 pwr.edu.pl MX
; (1 server found)
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 35306
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;pwr.edu.pl.                IN      MX

;; ANSWER SECTION:
pwr.edu.pl.                1872    IN      MX      10 alt3.aspmx.l.google.com.
pwr.edu.pl.                1872    IN      MX      5 alt1.aspmx.l.google.com.
pwr.edu.pl.                1872    IN      MX      7 alt2.aspmx.l.google.com.
pwr.edu.pl.                1872    IN      MX      0 aspmx.l.google.com.
pwr.edu.pl.                1872    IN      MX      15 alt4.aspmx.l.google.com.

;; Query time: 23 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Fri Dec 12 20:22:45 CET 2025
;; MSG SIZE rcvd: 157
```


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Reveals **mail infrastructure** (Google mail servers in this case). OSINT use: useful for **phishing, social engineering, or mail server enumeration**.

Run: > dig -x 156.17.75.200 | tee dig_156_17_75_200.txt

```
(stud@kali-vm)-[~]
$ dig -x 156.17.75.200 | tee dig_156_17_75_200.txt

; <<>> DiG 9.19.21-1+b1-Debian <<>> -x 156.17.75.200
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 39846
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 2, ADDITIONAL: 3

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: d182dd7fe34de397598f6960693c6c38e98460ac8a6f9f12 (good)
;; QUESTION SECTION:
;200.75.17.156.in-addr.arpa.      IN      PTR

;; ANSWER SECTION:
200.75.17.156.in-addr.arpa. 3536 IN      PTR      host-156-17-75-200.pwr.edu.pl.

;; AUTHORITY SECTION:
75.17.156.in-addr.arpa. 86336 IN      NS       dns2.pwr.edu.pl.
75.17.156.in-addr.arpa. 86336 IN      NS       dns.pwr.edu.pl.

;; ADDITIONAL SECTION:
dns.pwr.edu.pl.           301 IN      A        156.17.18.10
dns2.pwr.edu.pl.          301 IN      A        156.17.18.11

;; Query time: 0 msec
;; SERVER: 10.138.0.1#53(10.138.0.1) (UDP)
;; WHEN: Fri Dec 12 20:25:43 CET 2025
;; MSG SIZE rcvd: 195
```

Reverse DNS lookup links IP -> hostname. OSINT use: can **map hosts to domain names**, verify IP ownership, or find subdomains.

Run: > dig pwr.edu.pl +noall +answer | tee dig_noall_answer_pwr.txt

```
(stud@kali-vm)-[~]
$ dig pwr.edu.pl +noall +answer | tee dig_noall_answer_pwr.txt
pwr.edu.pl.           4291 IN      A        156.17.75.200
```

Quick, clean info on **live IPs** and TTL. OSINT use: assess **live targets**, caching, and network exposure for reconnaissance.

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Run: > dig pwr.edu.pl +trace | tee dig_trace_pwr.txt

```
(stud@kali-vm)-[~]
$ dig pwr.edu.pl +trace | tee dig_trace_pwr.txt

; <<>> DiG 9.19.21-1+b1-Debian <<>> pwr.edu.pl +trace
;; global options: +cmd
.      455485 IN      NS      h.root-servers.net.
.      455485 IN      NS      f.root-servers.net.
.      455485 IN      NS      k.root-servers.net.
.      455485 IN      NS      m.root-servers.net.
.      455485 IN      NS      g.root-servers.net.
.      455485 IN      NS      l.root-servers.net.
.      455485 IN      NS      c.root-servers.net.
.      455485 IN      NS      i.root-servers.net.
.      455485 IN      NS      j.root-servers.net.
.      455485 IN      NS      b.root-servers.net.
.      455485 IN      NS      e.root-servers.net.
.      455485 IN      NS      d.root-servers.net.
.      455485 IN      NS      a.root-servers.net.
.      455485 IN      RRSIG   NS 8 0 518400 20251224210000 20251211200000 61809 . eCiWEcNRjIFl00E
1GLH/NRi4IQxpID/XVDzv9g8fcWBN3wORGk7ANtc D9bqIijvL/+zpcXSWeSSvdwPh9hZ7Leg/3uIAHb7QxjTVRWKrMzOSjhj /RsKJpMZMvXlUrm8
+TRVklpLYCiPp9VIAG50r8G12IAigt8o37shY6tm CR31P2q0ZL625vFeXKzWxISdMaj66eylJB8YF7knGv2Q+SkLG2BC2ZJb Xtb1H66MrwbzGGC+s
FxiPn5PeAX2YZ8IINSnoyEJ+RZb3wxwgeZDsJb 1gX6RSXiYnLAXEiC5uCM9Nbyji+bnwHS7TSSEzj4tJ8N4jhotHwYeqY 1/noZA=
;; Received 1125 bytes from 10.138.0.1#53(10.138.0.1) in 4 ms

pl.      172800 IN      NS      f-dns.pl.
pl.      172800 IN      NS      b-dns.pl.
pl.      172800 IN      NS      j-dns.pl.
pl.      172800 IN      NS      h-dns.pl.
pl.      172800 IN      NS      d-dns.pl.
pl.      172800 IN      NS      a-dns.pl.
pl.      86400 IN      DS      35456 8 2 C1FCC88C73CCBAFAA2D6D68575B01ED753786CD733A4A239324D5609
7F17930B
pl.      86400 IN      RRSIG   DS 8 1 86400 20251225170000 20251212160000 61809 . rdSZqqiigNoKw6xL
14V5FVeHunJ01tCCEnXkQTWkZ3kBe0J+n6ZG0Ao/X pXkjz/66M569x9He3qrUsS36roRuS9BSdtILMvw3uFXx/Qpf4KIFrk9w Q5mk0cTEBeo2t4+iV
CDe/rXrGe0+zXjqYkINEFTqNafHQSwfVsCLspOF 54v01UpJrCnWIdlzhxENCQui+zDhvmBbqSEw6kSjbnWsuWUjKa+YvRqV igu0Vx5VMDfuAgOkCk
ZkMb7SAvsR6xblymih41gszc8KowH4gsTLFFg 6/DbGGIUp/LZoR3iKr/1SaLWQPRQsIzLG3ovqvYtVR02CmdjJDYfQSIz tJDjka=
;; Received 758 bytes from 198.41.0.4#53(a.root-servers.net) in 16 ms

pwr.edu.pl.      86400 IN      NS      dns3.pwr.edu.pl.
pwr.edu.pl.      86400 IN      NS      ns1.net.icm.edu.pl.
pwr.edu.pl.      86400 IN      NS      ns2.net.icm.edu.pl.
pwr.edu.pl.      86400 IN      NS      dns.pwr.edu.pl.
pwr.edu.pl.      86400 IN      NS      dns2.pwr.edu.pl.
pwr.edu.pl.      86400 IN      NS      sun2.pwr.wroc.pl.
SK240AE8RHU40P3HI9TS786POE3T266C.edu.pl. 3600 IN NSEC3 1 1 0 D99D35D740BC569E SL5Q188CDI9G1RCI18U0UOH0VVS01P8E NS S
OA RRSIG DNSKEY NSEC3PARAM
SK240AE8RHU40P3HI9TS786POE3T266C.edu.pl. 3600 IN RRSIG NSEC3 8 3 3600 20260106120000 20251207120000 32313 edu.pl. X
J6IMuDZIBKPPCUvk7PeM7gr/rvX5tkEEhZyWQEElP7oA4yGP8RiFqgR dp7A4nioCcalgWuTw6iWlZhenpLhQd/LyPgX4YCDAAQwgWao0wU0yC0uv d8
LCVGwOhobGMP7MrUPbigBD810scX+XRdAtqZa8pjfaonv9Z8N8dH/E UpYxZLfmFZx9lia9TWlyiM/U0qCU8NqdXqedwfmCTZ5RVD5mp+TZoh3J Dbk
TCaf2TSuSLNuaKIHdncwvdwWWQhoiHNWeTVF4HR9FuGZdHTWY30b ZmxdUnqt01ZPqQ95rBshSiquFk6upuZoqB96xAFsjK0Sn5cxw7jRJ4x pJNm
nG=
```

Shows **full delegation path** from root -> TLD -> authoritative NS. OSINT use: shows which NS control the domain, can identify **potential weak points or secondary NS**.

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- Find the primary DNS for a given domain

To find the authoritative DNS server of the university you need to run: > dig pwr.edu.pl SOA

```
(stud@kali-vm)-[~]
$ dig pwr.edu.pl SOA

; <<>> DiG 9.19.21-1+b1-Debian <<>> pwr.edu.pl SOA
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 44800
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 6, ADDITIONAL: 9

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: ad0d48dc5ec4765f30b61616693c6eea5f4987a1528f76ab (good)
;; QUESTION SECTION:
pwr.edu.pl.                IN      SOA

;; ANSWER SECTION:
pwr.edu.pl.                7200    IN      SOA      dns.pwr.edu.pl. dns.pwr.edu.pl. 2025121106 3600 300 1209600 7200

;; AUTHORITY SECTION:
pwr.edu.pl.                86268   IN      NS       dns.pwr.edu.pl.
pwr.edu.pl.                86268   IN      NS       dns3.pwr.edu.pl.
pwr.edu.pl.                86268   IN      NS       ns2.net.icm.edu.pl.
pwr.edu.pl.                86268   IN      NS       sun2.pwr.wroc.pl.
pwr.edu.pl.                86268   IN      NS       ns1.net.icm.edu.pl.
pwr.edu.pl.                86268   IN      NS       dns2.pwr.edu.pl.

;; ADDITIONAL SECTION:
dns.pwr.edu.pl.            7068    IN      A        156.17.18.10
ns1.net.icm.edu.pl.        56457   IN      A        213.135.57.10
ns2.net.icm.edu.pl.        70969   IN      A        193.219.28.51
dns2.pwr.edu.pl.           7068    IN      A        156.17.18.11
sun2.pwr.wroc.pl.          6613    IN      A        156.17.5.2
dns3.pwr.edu.pl.           7068    IN      A        156.17.197.98
ns1.net.icm.edu.pl.        56457   IN      AAAA     2001:6a0:0:2000::10
ns2.net.icm.edu.pl.        70969   IN      AAAA     2001:6a0:0:2002::51

;; Query time: 12 msec
;; SERVER: 10.138.0.1#53(10.138.0.1) (UDP)
;; WHEN: Fri Dec 12 20:37:13 CET 2025
;; MSG SIZE rcvd: 383
```

So the principal DNS server is: dns.pwr.edu.pl with:

- Serial: 2025121106 (YYYYMMDD of last update)
- Refresh: 3600 (sec = 1 hour)
- Retry: 300 (sec = 5 min)
- Expire: 1209600 (sec = 14 days)
- Min TTL: 7200 (Default caching time)

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- Try to find out what is TTL and if the requested domain was cached by DNS.

TTL (**Time To Live**) is the number of seconds a DNS record is allowed to stay in a DNS resolver's cache before it must ask the authoritative DNS server again.

- High TTL -> cached longer
- Low TTL -> refreshed more often

How to find it: when running dig pwr.edu.pl

```
(stud@kali-vm)-[~]
$ dig pwr.edu.pl

; <<>> DiG 9.19.21-1+b1-Debian <<>> pwr.edu.pl
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 19170
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 8, ADDITIONAL: 11

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: 790dc63af83245bb6d8a0ef7693c712e9d0fc7b69be609d3 (good)
;; QUESTION SECTION:
;pwr.edu.pl.                IN      A

;; ANSWER SECTION:
pwr.edu.pl.                 3274    IN      A      156.17.75.200

;; AUTHORITY SECTION:
pwr.edu.pl.                 85688   IN      NS      dns4.pwr.edu.pl.
pwr.edu.pl.                 85688   IN      NS      ns2.net.icm.edu.pl.
pwr.edu.pl.                 85688   IN      NS      ns1.net.icm.edu.pl.
pwr.edu.pl.                 85688   IN      NS      dns.pwr.edu.pl.
pwr.edu.pl.                 85688   IN      NS      dns.pwr.wroc.pl.
pwr.edu.pl.                 85688   IN      NS      dns2.pwr.edu.pl.
pwr.edu.pl.                 85688   IN      NS      sun2.pwr.wroc.pl.
pwr.edu.pl.                 85688   IN      NS      dns3.pwr.edu.pl.

;; ADDITIONAL SECTION:
dns.pwr.wroc.pl.            48672   IN      A      156.17.18.10
ns1.net.icm.edu.pl.         55877   IN      A      213.135.57.10
ns2.net.icm.edu.pl.         70389   IN      A      193.219.28.51
dns2.pwr.edu.pl.            6488    IN      A      156.17.18.11
dns.pwr.edu.pl.             6488    IN      A      156.17.18.10
dns3.pwr.edu.pl.            6488    IN      A      156.17.197.98
dns4.pwr.edu.pl.            6488    IN      A      156.17.140.4
sun2.pwr.wroc.pl.           6033    IN      A      156.17.5.2
ns1.net.icm.edu.pl.         55877   IN      AAAA    2001:6a0:0:2000::10
ns2.net.icm.edu.pl.         70389   IN      AAAA    2001:6a0:0:2002::51

;; Query time: 4 msec
;; SERVER: 10.138.0.1#53(10.138.0.1) (UDP)
;; WHEN: Fri Dec 12 20:46:53 CET 2025
;; MSG SIZE rcvd: 432
```

In the line: pwr.edu.pl. **3274** IN A 156.17.75.200

The observed TTL will keep going down until it reaches **0** and **another request is sent** and **caught again**.

For example, on the line: pwr.edu.pl. **85688** IN NS dns.pwr.wroc.pl.

85688 (sec) means that **it remains in cache memory for 23 more hours** until another request is sent.

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- Find out how long time ago the given domain was requested at some DNSs (e.g. 1.1.1.1, 8.8.8.8, local DNS,...)

You can write several commands: `> dig @x.x.x.x <dom ain>` or for local: `> dig <domain>`

```
(stud@kali-vm)-[~]
$ dig pwr.edu.pl

;<<>> DiG 9.19.21-1+b1-Debian <<>> pwr.edu.pl
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 37433
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 8, ADDITIONAL: 11

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: 89068bc986dbf804050d4dea693c73877d56e3a1987c8f2b (good)
;; QUESTION SECTION:
;pwr.edu.pl.                IN      A

;; ANSWER SECTION:
pwr.edu.pl.                2673    IN      A      156.17.75.200

;; AUTHORITY SECTION:
pwr.edu.pl.                85087   IN      NS      dns3.pwr.edu.pl.
pwr.edu.pl.                85087   IN      NS      dns.pwr.edu.pl.
pwr.edu.pl.                85087   IN      NS      dns2.pwr.edu.pl.
pwr.edu.pl.                85087   IN      NS      dns4.pwr.edu.pl.
pwr.edu.pl.                85087   IN      NS      ns1.net.icm.edu.pl.
pwr.edu.pl.                85087   IN      NS      dns.pwr.wroc.pl.
pwr.edu.pl.                85087   IN      NS      sun2.pwr.wroc.pl.
pwr.edu.pl.                85087   IN      NS      ns2.net.icm.edu.pl.

;; ADDITIONAL SECTION:
dns.pwr.wroc.pl.           48071   IN      A      156.17.18.10
ns1.net.icm.edu.pl.        55276   IN      A      213.135.57.10
ns2.net.icm.edu.pl.        69788   IN      A      193.219.28.51
dns2.pwr.edu.pl.           5887    IN      A      156.17.18.11
dns.pwr.edu.pl.            5887    IN      A      156.17.18.10
dns3.pwr.edu.pl.           5887    IN      A      156.17.197.98
dns4.pwr.edu.pl.           5887    IN      A      156.17.140.4
sun2.pwr.wroc.pl.          5432    IN      A      156.17.5.2
ns1.net.icm.edu.pl.        55276   IN      AAAA    2001:6a0:0:2000::10
ns2.net.icm.edu.pl.        69788   IN      AAAA    2001:6a0:0:2002::51

;; Query time: 4 msec
;; SERVER: 10.138.0.1#53(10.138.0.1) (UDP)
;; WHEN: Fri Dec 12 20:56:54 CET 2025
;; MSG SIZE rcvd: 432
```

```
(stud@kali-vm)-[~]
$ dig @8.8.8.8 pwr.edu.pl

;<<>> DiG 9.19.21-1+b1-Debian <<>> @8.8.8.8 pwr.edu.pl
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 19163
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;pwr.edu.pl.                IN      A

;; ANSWER SECTION:
pwr.edu.pl.                278     IN      A      156.17.75.200

;; Query time: 24 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Fri Dec 12 20:57:03 CET 2025
;; MSG SIZE rcvd: 55

(stud@kali-vm)-[~]
$ dig @1.1.1.1 pwr.edu.pl

;<<>> DiG 9.19.21-1+b1-Debian <<>> @1.1.1.1 pwr.edu.pl
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 16548
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;pwr.edu.pl.                IN      A

;; ANSWER SECTION:
pwr.edu.pl.                5062    IN      A      156.17.75.200

;; Query time: 16 msec
;; SERVER: 1.1.1.1#53(1.1.1.1) (UDP)
;; WHEN: Fri Dec 12 20:57:08 CET 2025
;; MSG SIZE rcvd: 55
```

As we know that: Authoritative TTL = 7200 seconds

The formula is $Cache\ age\ [sec] = 7200\ [sec] - Remaining\ TTL\ [sec]$

For Local DNS: $Cache\ age = 7200 - 2673 = 4527\ s \approx 1\ hour\ and\ 15\ minutes\ ago$

For 8.8.8.8 (Google): $Cache\ age = 7200 - 278 = 6922\ s \approx 1\ hour\ and\ 55\ minutes\ ago$

For 1.1.1.1 (Cloudflare): $Cache\ age = 7200 - 5062 = 2138\ s \approx 36\ minutes\ ago$

This is proof that **you are seeing cached answers**, not fresh ones.

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2.4 Utilize dnsenum dnsenum example.com dnsenum -f dns.txt example.com:

dnsenum performs multi-technique DNS enumeration: queries NS, MX, SOA, does brute-force subdomain lookups (wordlist based), reverse lookups on discovered nets, and attempts zone transfer tests.

To run it you must use the command: `> dnsenum pwr.edu.pl | tee dnsenum_pwr.txt`

```
(stud@kali-vm)-[~]
$ dnsenum pwr.edu.pl | tee dnsenum_pwr.txt
AXFR record query failed: REFUSED
AXFR record query failed: REFUSED
AXFR record query failed: REFUSED
AXFR record query failed: REFUSED
AXFR record query failed: REFUSED
AXFR record query failed: REFUSED
AXFR record query failed: REFUSED
AXFR record query failed: REFUSED
dnsenum VERSION:1.3.1

----- pwr.edu.pl -----
Host's addresses:

pwr.edu.pl.                5758      IN      A       156.17.75.200

Name Servers:

dns.pwr.edu.pl.            3030      IN      A       156.17.18.10
dns3.pwr.edu.pl.           3030      IN      A       156.17.197.98
ns2.net.icm.edu.pl.        11219     IN      A       193.219.28.51
dns4.pwr.edu.pl.           5758      IN      A       156.17.140.4
dns.pwr.wroc.pl.           5758      IN      A       156.17.18.10
ns1.net.icm.edu.pl.        82230     IN      A       213.135.57.10
dns2.pwr.edu.pl.           3030      IN      A       156.17.18.11
sun2.pwr.wroc.pl.          3030      IN      A       156.17.5.2

Mail (MX) Servers:

alt2.aspmx.l.google.com.   262       IN      A       192.178.163.27
alt4.aspmx.l.google.com.   262       IN      A       142.250.101.26
alt1.aspmx.l.google.com.   262       IN      A       108.177.125.27
aspmx.l.google.com.        262       IN      A       64.233.162.26
alt3.aspmx.l.google.com.   262       IN      A       172.217.78.27

Trying Zone Transfers and getting Bind Versions:
```

```
Trying Zone Transfer for pwr.edu.pl on sun2.pwr.wroc.pl ...
Trying Zone Transfer for pwr.edu.pl on dns4.pwr.edu.pl ...
Trying Zone Transfer for pwr.edu.pl on dns.pwr.wroc.pl ...
Trying Zone Transfer for pwr.edu.pl on dns2.pwr.edu.pl ...
Trying Zone Transfer for pwr.edu.pl on dns3.pwr.edu.pl ...
Trying Zone Transfer for pwr.edu.pl on ns2.net.icm.edu.pl ...
Trying Zone Transfer for pwr.edu.pl on ns1.net.icm.edu.pl ...
Trying Zone Transfer for pwr.edu.pl on dns.pwr.edu.pl ...
```

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Brute forcing with /usr/share/dnsenum/dns.txt:

autodiscover.pwr.edu.pl.	7200	IN	A	156.17.66.175
backup.pwr.edu.pl.	7200	IN	CNAME	di.pwr.wroc.pl.
di.pwr.wroc.pl.	7200	IN	A	10.25.0.12
beta.pwr.edu.pl.	7200	IN	CNAME	di.pwr.wroc.pl.
di.pwr.wroc.pl.	7200	IN	A	10.25.0.12
blog.pwr.edu.pl.	7200	IN	CNAME	di.pwr.wroc.pl.
di.pwr.wroc.pl.	7200	IN	A	10.25.0.12
di.pwr.edu.pl.	7200	IN	A	156.17.75.200
dns.pwr.edu.pl.	4469	IN	A	156.17.18.10
dns1.pwr.edu.pl.	7200	IN	A	156.17.140.3
dns2.pwr.edu.pl.	4469	IN	A	156.17.18.11
dns3.pwr.edu.pl.	4469	IN	A	156.17.197.98
e.pwr.edu.pl.	7200	IN	A	156.17.193.220
emc.pwr.edu.pl.	7200	IN	A	156.17.193.59
eportal.pwr.edu.pl.	2206	IN	A	156.17.70.219
forum.pwr.edu.pl.	7200	IN	A	156.17.75.210
ftp.pwr.edu.pl.	7200	IN	A	156.17.193.37
hermes.pwr.edu.pl.	7200	IN	A	156.17.75.200
ib.pwr.edu.pl.	7200	IN	A	156.17.193.59
intranet.pwr.edu.pl.	7200	IN	A	156.17.75.210
it.pwr.edu.pl.	7200	IN	CNAME	di.pwr.wroc.pl.
di.pwr.wroc.pl.	7197	IN	A	10.25.0.12
kbo.pwr.edu.pl.	7200	IN	A	156.17.75.200
live.pwr.edu.pl.	7200	IN	A	156.17.18.5
log.pwr.edu.pl.	7200	IN	A	156.17.18.129
mail.pwr.edu.pl.	7200	IN	A	156.17.193.179
ntp.pwr.edu.pl.	7200	IN	A	156.17.18.1
old.pwr.edu.pl.	7200	IN	A	156.17.193.247
portal.pwr.edu.pl.	7200	IN	A	156.17.75.200
repository.pwr.edu.pl.	7200	IN	A	156.17.79.16
sos.pwr.edu.pl.	7200	IN	A	156.17.66.16
ssc.pwr.edu.pl.	7200	IN	A	156.17.193.59
stats.pwr.edu.pl.	7200	IN	A	156.17.16.244
transport.pwr.edu.pl.	7200	IN	A	156.17.197.36
vpn.pwr.edu.pl.	7200	IN	A	156.17.33.70
www.pwr.edu.pl.	7200	IN	A	156.17.75.200

A = final IP

CNAME = pointer to another name

...

pwr.edu.pl ip blocks:

```
156.17.33.118/32
156.17.66.30/32
156.17.66.33/32
156.17.66.175/32
156.17.66.200/29
156.17.66.208/31
156.17.66.210/32
156.17.66.220/32
156.17.70.111/32
156.17.70.199/32
156.17.70.217/32
156.17.70.218/31
156.17.70.220/31
156.17.70.245/32
156.17.79.44/32
156.17.79.51/32
156.17.79.110/32
```

And run also the command: > dnsenum -f /usr/share/dnsenum/dns.txt pwr.edu.pl <domain> | tee <FileNameWhereToStore>

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That writes the hosts given dns.txt:

```
156.17.66.254/32
156.17.70.1/32
156.17.70.2/31
156.17.70.4/30
156.17.70.8/29
156.17.70.16/28
156.17.70.32/27
156.17.70.64/26
156.17.70.128/26
156.17.70.192/27
156.17.70.224/28
156.17.70.240/29
156.17.70.248/30
156.17.70.252/31
156.17.70.254/32
156.17.75.1/32
156.17.75.2/31
156.17.75.4/30
156.17.75.8/29
156.17.75.16/28
156.17.75.32/27
156.17.75.64/26
156.17.75.128/26
156.17.75.192/27
156.17.75.224/28
156.17.75.240/29
156.17.75.248/30
156.17.75.252/31
156.17.75.254/32
156.17.140.1/32
156.17.140.2/31
156.17.140.4/30
156.17.140.8/29
156.17.140.16/28
156.17.140.32/27
156.17.140.64/26
156.17.140.128/26
156.17.140.192/27
156.17.140.224/28
156.17.140.240/29
156.17.140.248/30
156.17.140.252/31
156.17.140.254/32
done.
```

With this attack we can identify additional parts of the domain infrastructure such as CNAME links and host IP addresses:

- autodiscover.pwr.edu.pl -> 156.17.66.175 ; www.pwr.edu.pl -> 156.17.75.200 (from log: dnsenum_f_pwr)
- Several hosts are resolved via the CNAME di.pwr.wroc.pl to an internal address 10.25.0.12, which indicates internal naming information leakage through DNS CNAME records (from log: dnsenum_f_pwr)
- Other publicly reachable services were identified, e.g. mail.pwr.edu.pl -> 156.17.193.179, ftp.pwr.edu.pl -> 156.17.193.37, vpn.pwr.edu.pl -> 156.17.33.70, etc.

This information allows mapping exposed services and understanding how internal and external DNS naming is structured, which is useful for further reconnaissance.

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2.5 Find all of the IP addresses and hostnames of a target fierce -dns example.com -threads 3:

Running the command: `> fierce -dns pwr.edu.pl -threads 3 | tee fierce_pwr.txt`, as we need the new version prompt we write `> fierce --domain pwr.edu.pl | tee fierce_pwr.txt`

What fierce does:

- Enumerates hostnames / subdomains via DNS.
- Detects parent netblocks and “nearby” addresses (hosts in the same subnets).
- Tests for wildcard DNS behavior and attempts zone transfers (AXFR).
- Produces candidate IP ranges to target with active scanning tools (Nmap, vulnerability scanners).

```
(stud@kali-vm)-[~]
$ fierce --domain pwr.edu.pl | tee fierce_pwr.txt
NS: dns.pwr.wroc.pl. ns2.net.icm.edu.pl. dns.pwr.edu.pl. dns2.pwr.edu.pl. dns3.pwr.edu.pl. sun2.pwr.wroc.pl. ns1.net.icm.edu.pl. dns4.pwr.edu.pl.
SOA: dns.pwr.edu.pl. (156.17.18.10)
Zone: failure
Wildcard: failure
Found: ad.pwr.edu.pl. (156.17.75.200)
Nearby:
{'156.17.75.195': 'host-156-17-75-195.pwr.edu.pl.',
'156.17.75.196': 'host-156-17-75-196.pwr.edu.pl.',
'156.17.75.197': 'host-156-17-75-197.pwr.edu.pl.',
'156.17.75.198': 'host-156-17-75-198.pwr.edu.pl.',
'156.17.75.199': 'host-156-17-75-199.pwr.edu.pl.',
'156.17.75.200': 'host-156-17-75-200.pwr.edu.pl.',
'156.17.75.201': 'host-156-17-75-201.pwr.edu.pl.',
'156.17.75.202': 'host-156-17-75-202.pwr.edu.pl.',
'156.17.75.203': 'host-156-17-75-203.pwr.edu.pl.',
'156.17.75.204': 'host-156-17-75-204.pwr.edu.pl.',
'156.17.75.205': 'host-156-17-75-205.pwr.edu.pl.'}
Found: ae.pwr.edu.pl. (156.17.193.59)
Nearby:
{'156.17.193.59': 'vm-webmin2.wcss.wroc.pl.',
'156.17.193.62': 'gw-v119.wask.wroc.pl.'}
Found: ai.pwr.edu.pl. (104.198.14.52)
Nearby:
{'104.198.14.47': '47.14.198.104.bc.googleusercontent.com.',
'104.198.14.48': '48.14.198.104.bc.googleusercontent.com.',
'104.198.14.49': '49.14.198.104.bc.googleusercontent.com.',
'104.198.14.50': '50.14.198.104.bc.googleusercontent.com.',
'104.198.14.51': '51.14.198.104.bc.googleusercontent.com.',
'104.198.14.52': '52.14.198.104.bc.googleusercontent.com.',
'104.198.14.53': '53.14.198.104.bc.googleusercontent.com.',
'104.198.14.54': '54.14.198.104.bc.googleusercontent.com.',
'104.198.14.55': '55.14.198.104.bc.googleusercontent.com.',
'104.198.14.56': '56.14.198.104.bc.googleusercontent.com.',
'104.198.14.57': '57.14.198.104.bc.googleusercontent.com.'}
Found: backup.pwr.edu.pl. (10.25.0.12)
Found: bb.pwr.edu.pl. (156.17.193.59)
Found: beta.pwr.edu.pl. (10.25.0.12)
Found: bg.pwr.edu.pl. (156.17.79.6)
Nearby:
{'156.17.79.2': 'proxy.bg.pwr.wroc.pl.',
'156.17.79.3': 'pliki.bg.pwr.wroc.pl.',
'156.17.79.4': 'smtp.bg.pwr.wroc.pl.',
'156.17.79.6': 'd21svw-ndns4.bg.pwr.wroc.pl.'}
Found: bi.pwr.edu.pl. (10.25.0.12)
Found: blog.pwr.edu.pl. (10.25.0.12)
Found: cc.pwr.edu.pl. (156.17.48.211)
```

...

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```
'156.17.193.181': 'vlan109-f5-local-szpadka.wcss.wroc.pl.',
'156.17.193.182': 'vlan109-f5-float.wcss.wroc.pl.',
'156.17.193.183': 'mail.pwr.wroc.pl.',
'156.17.193.184': 'mail.wcss.wroc.pl.}'
Found: me.pwr.edu.pl. (156.17.193.59)
Found: monitor.pwr.edu.pl. (10.25.0.12)
Found: monitoring.pwr.edu.pl. (10.25.0.12)
Found: ms.pwr.edu.pl. (156.17.8.14)
Nearby:
{'156.17.8.14': 'io.immt.pwr.wroc.pl.',
'156.17.8.19': 'DIR-615.immt.pwr.wroc.pl.}'
Found: multimedia.pwr.edu.pl. (156.17.193.247)
Nearby:
{'156.17.193.242': 'adresy.pwr.wroc.pl.',
'156.17.193.244': 'uniwirt3.pwr.wroc.pl.',
'156.17.193.246': 'uniwirt.pwr.wroc.pl.',
'156.17.193.248': 'www.wcss.pl.',
'156.17.193.251': 'vlan129-f5-local-mieczyk.wcss.wroc.pl.}'
Found: mysql.pwr.edu.pl. (156.17.193.237)
Nearby:
{'156.17.193.232': 'upload.e-science.pl.',
'156.17.193.234': 'kangur.wcss.pl.',
'156.17.193.235': 'kalendarz.wcss.pl.',
'156.17.193.236': 'mysql.wcss.wroc.pl.',
'156.17.193.237': 'mysql.pwr.edu.pl.',
'156.17.193.240': 'oracle-0.wcss.wroc.pl.',
'156.17.193.241': 'epos.wcss.pl.}'
Found: na.pwr.edu.pl. (10.25.0.12)
Found: nas.pwr.edu.pl. (10.25.0.12)
Found: ntp.pwr.edu.pl. (156.17.18.1)
Found: old.pwr.edu.pl. (156.17.193.247)
Found: open.pwr.edu.pl. (156.17.70.100)
Nearby:
{'156.17.70.100': 'host-156-17-70-100.pwr.edu.pl.',
'156.17.70.101': 'host-156-17-70-101.pwr.edu.pl.',
'156.17.70.102': 'host-156-17-70-102.pwr.edu.pl.',
'156.17.70.103': 'host-156-17-70-103.pwr.edu.pl.',
'156.17.70.104': 'host-156-17-70-104.pwr.edu.pl.',
'156.17.70.105': 'host-156-17-70-105.pwr.edu.pl.',
'156.17.70.95': 'host-156-17-70-95.pwr.edu.pl.',
'156.17.70.96': 'host-156-17-70-96.pwr.edu.pl.',
'156.17.70.97': 'host-156-17-70-97.pwr.edu.pl.',
'156.17.70.98': 'host-156-17-70-98.pwr.edu.pl.',
'156.17.70.99': 'host-156-17-70-99.pwr.edu.pl.}'
Found: pf.pwr.edu.pl. (10.250.0.177)
Found: pki.pwr.edu.pl. (10.25.0.12)
Found: portal.pwr.edu.pl. (156.17.75.200)
Found: ppp.pwr.edu.pl. (156.17.75.200)
Found: ps.pwr.edu.pl. (156.17.75.200)
```

The scan revealed authoritative DNS servers (e.g. **dns.pwr.edu.pl**, **ns1.net.icm.edu.pl**), confirmed **AXFR** and **wildcard DNS failures**, and enumerated many subdomains with IPs (e.g. **ad.pwr.edu.pl** -> **156.17.75.200**, **ai.pwr.edu.pl** -> **104.198.14.52**).

It also identified multiple public netblocks (e.g. **156.17.75.***, **156.17.193.***) and exposed private/internal addresses in DNS (e.g. **backup.pwr.edu.pl** -> **10.25.0.12**, **guest.pwr.edu.pl** -> **10.250.13.254**).

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2.6 Get network routing information:

A. Using tcptraceroute traceroute www.example.com

Run the command: `> sudo tcptraceroute pwr.edu.pl | tee tcptraceroute_pwr.txt`

```
(stud@kali-vm)-[~]
$ sudo tcptraceroute pwr.edu.pl | tee tcptraceroute_pwr.txt
Running:
      traceroute -T -O info pwr.edu.pl
traceroute to pwr.edu.pl (156.17.75.200), 30 hops max, 60 byte packets
 1  host-156-17-75-200.pwr.edu.pl (156.17.75.200) <syn,ack,mss=1460>  5.679 ms  5.441 ms  7.268 ms
```

The trace reached **pwr.edu.pl (156.17.75.200)** in **one hop**, with a TCP **SYN/ACK** response and low latency (~5–7 ms). This indicates that the destination is either on the same local network (I am not on it) or that **intermediate routers/firewalls do not return TTL-exceeded messages for TCP probes**, causing only the final host to appear in the trace.

B. Using tctrace tctrace -i <network interface> -d <target host>

After checking the internet network in my case eth0: given the command `> ip a`

Run the command: `sudo tctrace -i eth0 -d pwr.edu.pl | tee tctrace_pwr.txt`

```
(stud@kali-vm)-[~]
$ sudo tctrace -i eth0 -d pwr.edu.pl | tee tctrace_pwr.txt
1(1)  [156.17.75.200] (reached; open)
```

The output shows the target **156.17.75.200** as **reached; open** in a single hop. This confirms successful TCP connectivity to the destination and that the probed TCP port is open, while intermediate routing devices do not expose their hop information.

2.7 Summarize your findings, compare tools and obtained results. What type of information do they provide? How malicious user can benefit from this type of information? Which tool is the most versatile?

What we got

- **DNS & hosting:** authoritative nameservers, SOA and TTLs confirmed; public A records show the main IPs for pwr.edu.pl.
- **Public hosts / subdomains found:** web, mail, vpn, ftp and many other subdomains were enumerated (a ready map of public-facing services).
- **Information leakage:** some DNS records (CNAMEs) reveal internal names / private Ips, accidental exposure of internal structure.

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- **Reachability:** several discovered hosts are directly reachable from the Internet (low latency, respond to TCP probes).
- **OSINT:** public emails and documents indexed on the web were collected (useful for profiling and phishing).
- **Main risks:** leaked internal names, multiple externally exposed services, and harvestable emails/documents that enable social engineering.

Tools vs results (what each tool gave us):

- **dig / host** -> authoritative DNS records (A, MX, NS, SOA, TTLs).
Good for facts about domain delegation and TTLs.
- **dnsenum / fierce** -> large list of subdomains, reverse lookups, AXFR checks, and discovery of CNAMEs that leak internal names.
- **tcptraceroute / tctrace** -> **reachability** and rough **network** path/latency for specific TCP services.
- **theHarvester** -> **harvested email addresses**, indexed documents and other publicly exposed content useful for profiling.
- **Shodan / Censys (workflow step)** -> how to check which services/ports are exposed globally and whether known vulnerable versions appear.

What info these tools provide, how it helps attackers *and* defenders, and the most versatile tool:

The tools deliver three main kinds of information: **(1)** structural facts about the domain (nameservers, TTLs, delegation), **(2)** an inventory of public hosts/services (subdomains, IPs, open TCP services), and **(3)** people/content intelligence (emails, indexed docs).

- A **malicious user** can use that information to build a target list, craft phishing/spear-phishing, look for exposed services with known CVEs, and map internal naming conventions.
- A **defender** uses the same outputs to find accidental leaks (fix CNAMEs), reduce unnecessary public exposure, prioritise patching (by mapping service versions -> CVEs), and to harden access (bastions, IP restrictions, MFA, remove public docs).

The most versatile tool from our work is **dnsenum** because it combines many DNS techniques (subdomain discovery, reverse lookups, AXFR tests) and produces the richest, immediately actionable map of hosts. So, it is great for both comprehensive reconnaissance and for defenders doing an exposure audit.

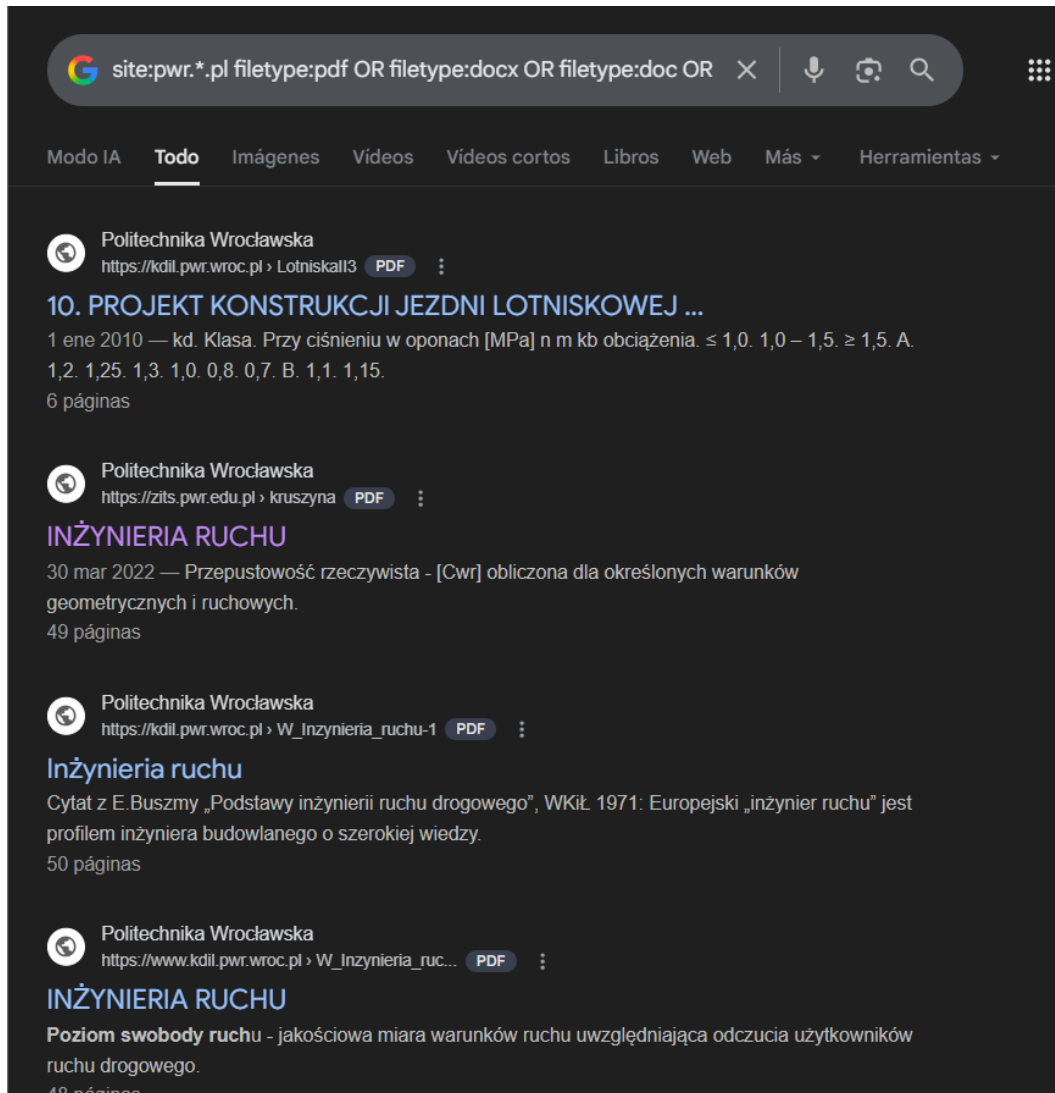
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2.8 Try to find some Vulnerable Files or sensible information using google hacking method (e.g. in domain pwr.*.edu.pl or in some other domain):

A. Search for documents files (.doc, .docx, .txt, .xls, pdf,....)

For that you can add the query: > site:pwr.*.pl filetype:pdf OR filetype:docx OR filetype:doc OR filetype:xlsx OR filetype:xls OR filetype:txt

Into the browser so it shows you any file given that specific format.



It is used to locate publicly accessible documents.

These searches returned **lecture notes, administrative documents, presentations, and reports**, some of which were not intended for broad public access.

There are several examples: “INŻYNIERIA RUCHU”.pdf, or “wytyczne do projektowania pomieszczeń”

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B. Search for documents with some specific content (e.g. grade, password, points, addresses, ...)

You can in this part expand what was told before with: `> site:pwr.*.pl filetype:pdf grade`

The screenshot shows Google search results for the query `site:pwr.*.pl filetype:pdf grade`. The results list several PDF documents from Politechnika Wrocławska. The first result is 'Internal assessment criteria—SL and HL' from <https://alfa.im.pwr.edu.pl/CriteriaAA2021>. The second is 'Algebra, Mathematical Analysis 1 & 2, 30h of lectures + ...' from <https://alfa.im.pwr.edu.pl/~michalik>. The third is 'Course credit regulations' from <https://alfa.im.pwr.edu.pl/~loch/credits>. The fourth is 'algebra and analytic geometry' from <https://alfa.im.pwr.edu.pl/dydaktyka/Karty>. The fifth is 'Excellent. It will contain advanc' from <https://alfa.im.pwr.edu.pl/Grade71IB2sem2020>.

Statystyka i rachunek prawdopodobieństwa

Wykład 3

Przykład

W pewnej firmie wylosowano próbkę $n = 30$ pracowników i spytano o wysokość ich wynagrodzenia netto. Uzyskano następujące wyniki (w tys. zł):

nr obserwacji	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
wartość	4,8	5,2	3,5	5,8	5,2	5,3	4,8	6,2	5,5	5,6	5,1	2,9	4,1	2,8	4,2

nr obserwacji	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
wartość	4,9	3,8	4,4	3,2	2,4	3,1	5,6	4,7	4,3	2,5	6,8	4,2	3,4	3,7	4,3

Zmienna losowa **ilościowa** X opisująca wysokość zarobków ma **rozkład ciągły**, gdyż teoretycznie może przyjmować dowolne wartości z określonego przedziału. Analizę powyższych danych rozpoczynamy od zbudowania **szeregu rozdzielczego**. Najpierw ustalamy liczbę klas np. $k = 5$ (slajd nr 10). Najmniejsza zaobserwowana wartość to 2,4 tys. a największa to 6,8 tys. Wobec tego przedział $[2,7]$ dzielimy na 5 jednakowych przedziałów: $[2,3)$, $[3,4)$, $[4,5)$, $[5,6)$, $[6,7)$, a następnie obliczamy ile wartości wpada do każdego z przedziałów. Otrzymujemy następującą tabelę, którą nazywamy **szeregiem rozdzielczym** (slajd nr 8):

przedział	[2,3)	[3,4)	[4,5)	[5,6)	[6,7)
liczność	4	6	10	8	2
częstość	0,133	0,200	0,333	0,267	0,067

Where you can see files as “Statystyka i rachunek prawdopodobieństwa”

Or with `site:pwr.*.pl address` or `site:pwr.*.pl password`

The screenshot shows Google search results for the query `site:pwr.*.pl address`. The results list several documents from Politechnika Wrocławska. The first is 'Słowość' from <http://plwordnet.pwr.wroc.pl/w...>. The second is 'address 8 social skill' from <http://plwordnet.pwr.wroc.pl/w...>. The third is 'Praca dyplomowa - wymagania' from <https://wii.pwr.edu.pl/studenci>. The fourth is 'Tomasz Żórawik: home page' from <https://alfa.im.pwr.edu.pl/~zora...>. The fifth is 'Karol Marchwicki Teaching' from <https://alfa.im.pwr.edu.pl/teaching>.

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C. Create statistics of findings (number of files, number of IP addresses, number of users, ...)

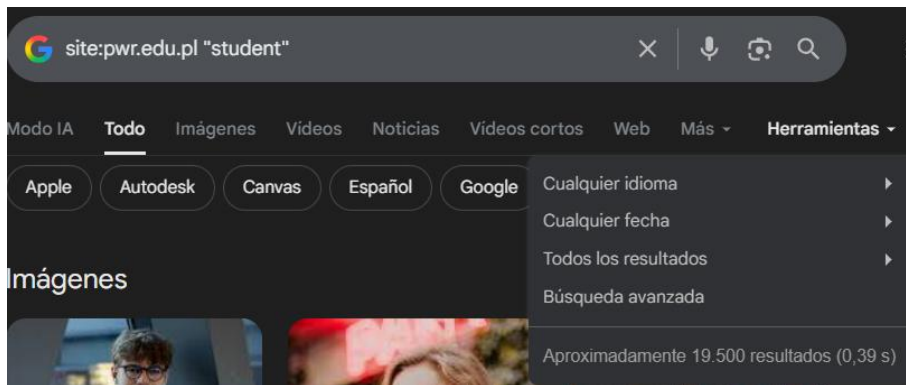
For the site: pwr.edu.pl with a total number of 509 000 results, from those, when filtering by type:

filetype:	pdf	doc	docx	xls	xlsx	txt
# of results	61100	1600	3630	241	678	462

Files containing the keywords:

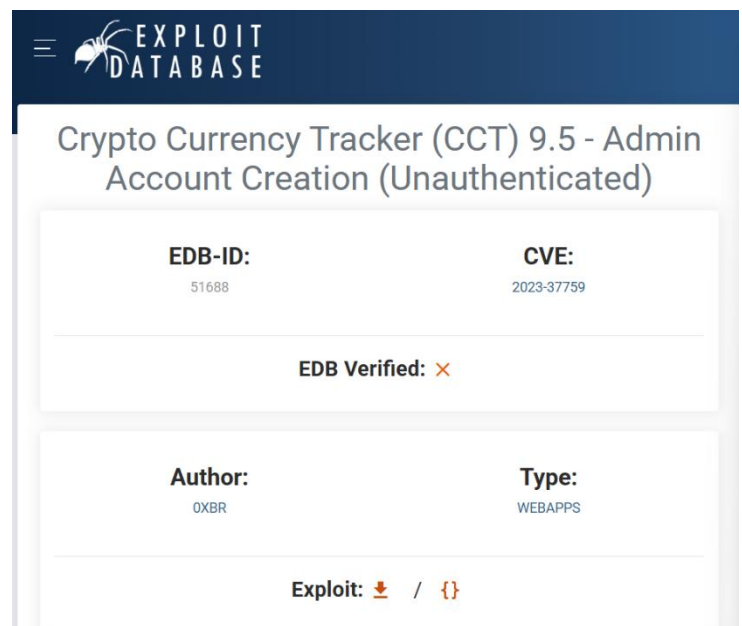
Keyword	"password"	"grade"	"exam"	"address"	"student"	"login"
# of results	909	2890	3420	6660	19500	11600

To check all the numbers you click on "Herramientas" (Tools) and it says 19.500 aprox.



D. Find some interesting data using selected dorks from <https://www.exploit-db.com/google-hacking-database/>

The Google Hacking Database (GHDB) on Exploit-DB contains numerous dorks that can uncover sensitive data. For example, `intitle:"index of" "parent directory"` reveals open directories with potentially sensitive files, while `inurl:admin` identifies unprotected admin login pages. Dorks like `filetype:log "password="` can expose log files containing passwords, and `inurl:"server-status"` finds exposed Apache server-status pages revealing server details.



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2.9 Use theharvester to collect e-mail accounts, username, and hostname/subdomains: theharvester -d example.com -l 100 -b linkedin for a given domain name try to search data using different data sources (e.g. baidu, bing, yahoo or use all):

To use theHarvester we need to use this form: theHarvester -d <domain> -l <limit> -b <source>:

Where domain is pwr.edu.pl, the limit is 200 results per source, and the sources are:

Domain	bing	duckduckgo	baidu	yahoo	otx	certspotter
IPs	-	-	-	-	-	-
Emails	-	-	10	7	-	-
Hosts	-	30	5	10	-	1

Domain	anubis	rapiddns	sitedossier	subdomaincenter	subdomainfinderc99	threatminer
IPs	-	-	-	-	-	-
Emails	-	-	-	-	-	-
Hosts	-	2234	-	489	-	-

Domain	dnsdumpster	urlscan	hackertarget	crtsh
IPs	-	19 IPs	-	-
		45 URLs		
Emails	-	-	-	-
Hosts	-	34	934	-

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```
[*] Target: pwr.edu.pl
[*] Searching Duckduckgo.
[*] No IPs found.
[*] No emails found.
[*] Hosts found: 30

2Fcrm.pwr.edu.pl
biblioteka.pwr.edu.pl
biurokarier.pwr.edu.pl
biznes.pwr.edu.pl
crm.pwr.edu.pl
dostepnosc.pwr.edu.pl
dsm.pwr.edu.pl
dwm.pwr.edu.pl
inkubator.pwr.edu.pl
jelenia-gora.pwr.edu.pl
konferencje.pwr.edu.pl
legnica.pwr.edu.pl
rekrutacja.pwr.edu.pl
sjo.pwr.edu.pl
summerschools.pwr.edu.pl
wa.pwr.edu.pl
walbrzych.pwr.edu.pl
wbliw.pwr.edu.pl
wch.pwr.edu.pl
wefim.pwr.edu.pl
weny.pwr.edu.pl
wggg.pwr.edu.pl
wis.pwr.edu.pl
wit.pwr.edu.pl
wm.pwr.edu.pl
wmat.pwr.edu.pl
wme.pwr.edu.pl
wmed.pwr.edu.pl
wppt.pwr.edu.pl
wz.pwr.edu.pl
```

```
[*] Target: pwr.edu.pl
[*] Searching Yahoo.
[*] No IPs found.
[*] Emails found: 7

dziekanat.wit@pwr.edu.pl
jaroslaw.michalek@pwr.edu.pl
michal.musial@pwr.edu.pl
rekrutacja@pwr.edu.pl
tomasz.trapko@pwr.edu.pl
wydz.bud@pwr.edu.pl
xxxxxx@student.pwr.edu.pl

[*] Hosts found: 10

di.pwr.edu.pl
eportal.pwr.edu.pl
poczta.student.pwr.edu.pl
registration.pwr.edu.pl
rekrutacja.pwr.edu.pl
student.pwr.edu.pl
wa.pwr.edu.pl
wbliw.pwr.edu.pl
wit.pwr.edu.pl
wz.pwr.edu.pl
```

```
[*] Target: pwr.edu.pl
[*] Searching Rapiddns.
[*] No IPs found.
[*] No emails found.
[*] Hosts found: 2234

02.tenere.konar.pwr.edu.pl:tenere.konar.pwr.edu.pl
02.tenere.konar.pwr.edu.pl:tenere.konar.pwr.edu.pl.
156-17-130-100.ii.pwr.edu.pl:156.17.130.100
156-17-130-101.ii.pwr.edu.pl:156.17.130.101
156-17-130-102.ii.pwr.edu.pl:156.17.130.102
156-17-130-103.ii.pwr.edu.pl:156.17.130.103
156-17-130-103.ii.pwr.edu.pl:64:ff9b::9c11:8267
156-17-130-104.ii.pwr.edu.pl:156.17.130.104
156-17-130-105.ii.pwr.edu.pl:156.17.130.105
156-17-130-106.ii.pwr.edu.pl:156.17.130.106
156-17-130-108.ii.pwr.edu.pl:156.17.130.108
156-17-130-109.ii.pwr.edu.pl:156.17.130.109
156-17-130-110.ii.pwr.edu.pl:156.17.130.110
156-17-130-114.ii.pwr.edu.pl:156.17.130.114
156-17-130-115.ii.pwr.edu.pl:156.17.130.115
156-17-130-116.ii.pwr.edu.pl:156.17.130.116
156-17-130-118.ii.pwr.edu.pl:156.17.130.118
156-17-130-118.ii.pwr.edu.pl:64:ff9b::9c11:8276
156-17-130-119.ii.pwr.edu.pl:156.17.130.119
156-17-130-119.ii.pwr.edu.pl:64:ff9b::9c11:8277
156-17-130-121.ii.pwr.edu.pl:156.17.130.121
156-17-130-122.ii.pwr.edu.pl:156.17.130.122
156-17-130-123.ii.pwr.edu.pl:156.17.130.123
156-17-130-125.ii.pwr.edu.pl:156.17.130.125
156-17-130-125.ii.pwr.edu.pl:64:ff9b::9c11:827d
156-17-130-126.ii.pwr.edu.pl:156.17.130.126
156-17-130-13.ii.pwr.edu.pl:156.17.130.13
156-17-130-132.ii.pwr.edu.pl:156.17.130.132
156-17-130-132.ii.pwr.edu.pl:64:ff9b::9c11:8284
156-17-130-133.ii.pwr.edu.pl:156.17.130.133
156-17-130-133.ii.pwr.edu.pl:64:ff9b::9c11:8285
156-17-130-134.ii.pwr.edu.pl:156.17.130.134
156-17-130-134.ii.pwr.edu.pl:64:ff9b::9c11:8286
156-17-130-135.ii.pwr.edu.pl:156.17.130.135
156-17-130-136.ii.pwr.edu.pl:156.17.130.136
156-17-130-136.ii.pwr.edu.pl:64:ff9b::9c11:8288
156-17-130-137.ii.pwr.edu.pl:156.17.130.137
156-17-130-137.ii.pwr.edu.pl:64:ff9b::9c11:8289
156-17-130-138.ii.pwr.edu.pl:156.17.130.138
```

```
[*] Target: pwr.edu.pl

An exception has occurred: Server disconnected
An exception has occurred: Server disconnected
An exception has occurred: Server disconnected
An exception has occurred: Server disconnected
An exception has occurred: Server disconnected
An exception has occurred: Server disconnected
An exception has occurred: Server disconnected
[*] Searching Baidu.

[*] No IPs found.

[*] Emails found: 10

266553@student.pwr.edu.pl
arkadiusz.lipiecki@pwr.edu.pl
bartosz.uniejewski@pwr.edu.pl
karol.szczypkowski@pwr.edu.pl
katarzyna.fojciak@pwr.edu.pl
katarzyna.maciejowska@pwr.edu.pl
malgorzata.skowronek@pwr.edu.pl
michal.gawor@pwr.edu.pl
milosz.rybak@pwr.edu.pl
pawel.ksieniewicz@pwr.edu.pl

[*] Hosts found: 5

aciids.pwr.edu.pl
opticaapplicata.pwr.edu.pl
osn.pwr.edu.pl
prac.im.pwr.edu.pl
student.pwr.edu.pl
```

```
[*] Target: pwr.edu.pl

Searching results.
[*] Searching Certspotter.

[*] No IPs found.

[*] No emails found.

[*] Hosts found: 1

*.pwr.edu.pl
```


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```
[*] Target: pwr.edu.pl

[*] Searching Subdomaincenter.

[*] No IPs found.

[*] No emails found.

[*] Hosts found: 489
```

```
10domowa.pwr.edu.pl
156-17-130-101o.ii.pwr.edu.pl
156-17-130-102o.ii.pwr.edu.pl
156-17-130-108o.ii.pwr.edu.pl
156-17-130-109o.ii.pwr.edu.pl
156-17-130-110o.ii.pwr.edu.pl
156-17-130-115o.ii.pwr.edu.pl
156-17-130-119o.ii.pwr.edu.pl
156-17-130-121o.ii.pwr.edu.pl
156-17-130-125o.ii.pwr.edu.pl
156-17-130-132m.ii.pwr.edu.pl
156-17-130-134o.ii.pwr.edu.pl
156-17-130-138o.ii.pwr.edu.pl
156-17-130-146o.ii.pwr.edu.pl
156-17-130-154o.ii.pwr.edu.pl
156-17-130-159o.ii.pwr.edu.pl
156-17-130-164o.ii.pwr.edu.pl
156-17-130-167o.ii.pwr.edu.pl
156-17-130-178o.ii.pwr.edu.pl
156-17-130-181o.ii.pwr.edu.pl
156-17-130-184o.ii.pwr.edu.pl
156-17-130-191o.ii.pwr.edu.pl
156-17-130-193o.ii.pwr.edu.pl
156-17-130-197o.ii.pwr.edu.pl
156-17-130-201o.ii.pwr.edu.pl
156-17-130-211o.ii.pwr.edu.pl
156-17-130-21o.ii.pwr.edu.pl
156-17-130-222o.ii.pwr.edu.pl
156-17-130-230m.ii.pwr.edu.pl
156-17-130-231o.ii.pwr.edu.pl
156-17-130-232o.ii.pwr.edu.pl
156-17-130-233o.ii.pwr.edu.pl
156-17-130-238o.ii.pwr.edu.pl
156-17-130-240m.ii.pwr.edu.pl
156-17-130-245o.ii.pwr.edu.pl
156-17-130-246o.ii.pwr.edu.pl
156-17-130-35m.ii.pwr.edu.pl
156-17-130-44m.ii.pwr.edu.pl
```

```
[*] Target: pwr.edu.pl

[*] Searching Urlscan.

[*] ASNS found: 5
```

```
AS13335
AS396982
AS41798
AS54113
AS8970
```

```
[*] Interesting Urls found: 45
```

```
https://alive.pwr.edu.pl/
https://api.pwr.edu.pl/
https://architectus.pwr.edu.pl/
https://auth.konar.pwr.edu.pl/flow/default-auth
https://beltsplices.pwr.edu.pl/
https://craft.pwr.edu.pl/
https://dwas.pwr.edu.pl/
https://eportal.pwr.edu.pl/
https://han.bg.pwr.edu.pl/
https://historycznekamienice.pwr.edu.pl/
https://historycznekamienice.pwr.edu.pl/login.php
https://ises.pwr.edu.pl/cache/?sugar=TOTO12
https://kinf.pwr.edu.pl/
https://kkmp2023.pwr.edu.pl/
https://laboratorium-emisji.pwr.edu.pl/
https://login.pwr.edu.pl/auth/realms/pwr.edu.pl/protocol/openid-connect/auth?client_id=account-co
-https%3A%2F%2Flogin.pwr.edu.pl%2Fauth%2Frealms%2Fpwr.edu.pl%2Faccount%2Fresponse_type=code&code_ch
X3bHmuOk02Jyi0PxiwRe38VUWbwb--gc6code_challenge_method=S256
https://master.web360-test.portale.pwr.edu.pl/
https://mizar.ktp.pwr.edu.pl/
https://odrarivercup.pwr.edu.pl/
https://pkg.pwr.edu.pl/
https://polnor.pwr.edu.pl/
https://prac.im.pwr.edu.pl/~amarczak/
https://prs.pwr.edu.pl/?page_id=510
https://ptw.pwr.edu.pl/
https://pwr.edu.pl/
https://si.pwr.edu.pl/
https://skanowanie3d.pwr.edu.pl/
https://solvro.pwr.edu.pl/
https://solvro.pwr.edu.pl/pl/
https://ssa.pwr.edu.pl/
https://studiujchemie.pwr.edu.pl/
https://sympozjon.pwr.edu.pl/
https://uptime.konar.pwr.edu.pl/dashboard
https://vpn.tenere.konar.pwr.edu.pl/
```

```
[*] Target: pwr.edu.pl
```

```
[*] Searching Hackertarget.
```

```
[*] No IPs found.
```

```
[*] No emails found.
```

```
[*] Hosts found: 934
```

```
10domow.pwr.edu.pl:156.17.193.59
20kkmgig.pwr.edu.pl:156.17.193.3
225.dds.pwr.edu.pl:156.17.229.10
225.dds.pwr.edu.pl:156.17.229.225
227.dds.pwr.edu.pl:156.17.229.227
228.dds.pwr.edu.pl:156.17.229.228
234.dds.pwr.edu.pl:156.17.229.10
234.dds.pwr.edu.pl:156.17.229.234
235.dds.pwr.edu.pl:156.17.229.10
235.dds.pwr.edu.pl:156.17.229.235
44zfp.pwr.edu.pl:156.17.193.59
44zsmgig.pwr.edu.pl:156.17.193.59
70-lecie-wa.pwr.edu.pl:156.17.29.227
B4427PC15.WM.pwr.edu.pl:156.17.6.235
DESKTOP-RK093PU.WM.pwr.edu.pl:156.17.6.243
abaqus.b5.pwr.edu.pl:156.17.14.159
absolwent-old.pwr.edu.pl:156.17.193.58
absolwent.pwr.edu.pl:156.17.75.200
acc.pwr.edu.pl:156.17.197.122
accusol.pwr.edu.pl:156.17.193.59
aciids.pwr.edu.pl:156.17.193.58
acps.pwr.edu.pl:156.17.193.59
act-dir.pwr.edu.pl:156.17.193.59
actabio.pwr.edu.pl:156.17.75.200
ad.pwr.edu.pl:156.17.75.200
admin.dam.pwr.edu.pl:156.17.12.142
admission.pwr.edu.pl:156.17.75.200
adresy.pwr.edu.pl:156.17.193.242
adsender.pwr.edu.pl:156.17.70.199
aerospace.pwr.edu.pl:156.17.193.58
aes.pwr.edu.pl:156.17.193.59
agrobiotechconference.pwr.edu.pl:156.17.193.58
ahead.b5.pwr.edu.pl:156.17.14.239
ai.pwr.edu.pl:104.198.14.52
aken.pwr.edu.pl:156.17.75.200
akig.pwr.edu.pl:156.17.193.3
akl.pwr.edu.pl:156.17.193.59
aklstol.pwr.edu.pl:156.17.193.59
aksep.pwr.edu.pl:156.17.193.59
aktywni.pwr.edu.pl:156.17.197.25
```

```
https://www.historycznekamienice.pwr.edu.pl/
https://www.in4music.pwr.edu.pl/
https://www.laboratorium-emisji.pwr.edu.pl/
https://www.smog.pwr.edu.pl/
https://www.snaa.pwr.edu.pl/
https://www.ubezpieczeniaspoleczne.pwr.edu.pl/
https://www.umir.pwr.edu.pl/
https://www.witai.pwr.edu.pl/
https://zapisy.pwr.edu.pl/
https://zatriyb.wppt.pwr.edu.pl/
https://zneee.pwr.edu.pl/
```

```
[*] IPs found: 19
```

```
104.198.14.52
156.17.1.253
156.17.16.240
156.17.16.248
156.17.193.3
156.17.193.58
156.17.193.59
156.17.229.201
156.17.7.2
156.17.70.219
156.17.75.200
156.17.75.210
156.17.79.40
156.17.9.14
156.17.9.150
172.67.203.45
185.199.110.153
188.114.96.3
193.111.175.5
```

```
[*] No emails found.
```

```
[*] Hosts found: 34
```

```
alive.pwr.edu.pl
api.pwr.edu.pl
architectus.pwr.edu.pl
auth.konar.pwr.edu.pl
beltsplices.pwr.edu.pl
craft.pwr.edu.pl
dwas.pwr.edu.pl
eportal.pwr.edu.pl
han.bg.pwr.edu.pl
historycznekamienice.pwr.edu.pl
```


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2.10 Using <https://nvd.nist.gov/vuln/search> search for some vulnerabilities in some type of the service (ssh, ftp, ssl, apache, qnap, western digital ...) and in next query related to some device (e.g. wireless router, asus wireless router, tp-link). Find some specific problem related to this service (device) – it is described as CVE – year – number. Find critical vulnerabilities. Describe the vulnerability. Find the link to the webpage with exploit or/and detailed description of the vulnerability:

Search by service, example ssh critical:

NVD Vulnerability Search

ssh

For a phrase search, use " "

Keyword: ssh CVSS v3.0 Severity Range: Critical (9.0 - 10.0)

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Identifier	CISA Key Info	Published Date	CNA	Description
CVE-2025-6237		2025-09-18	huntr.dev	A vulnerability in invokeai version v6.0.0a1 and below allows attackers to perform path traversal and arbitrary file deletion via the GET /api/v1/images/download/{bulk_download_item_name} endpoint. By manipulating the filename arguments, attackers can read and delete any files on the server, includi...
CVE-2024-6829		2025-03-20	huntr.dev	A vulnerability in aimhubio/aim version 3.19.3 allows an attacker to exploit the `tarfile.extractall()` function to extract the contents of a maliciously crafted tarfile to arbitrary locations on the host server. The attacker can control `repo.path` and `run_hash` to bypass directory existence check...
CVE-2024-11042		2025-03-20	huntr.dev	In invoke-ai/invokeai version v5.0.2, the web API `POST /api/v1/images/delete` is vulnerable to Arbitrary File Deletion. This vulnerability allows unauthorized attackers to delete arbitrary files on the server, potentially including critical or sensitive system files such as SSH keys, SQLite databas...
CVE-2024-		2025-03-20	huntr.dev	eosphoros-ai/db-gpt version 0.6.0 contains a vulnerability in the RAG-knowledge endpoint that allows for arbitrary file write. The issue arises from the ability to pass an absolute path to a call to `os.path.join`, enabling an attacker to write files to

Or for type of device, example wireless router:

NVD Vulnerability Search

wireless router

For a phrase search, use " "

Keyword: wireless router CVSS v3.0 Severity Range: Critical (9.0 - 10.0)

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Identifier	CISA Key Info	Published Date	CNA	Description
CVE-2020-3331		2020-07-16	Cisco Systems, Inc.	A vulnerability in the web-based management interface of Cisco RV110W Wireless-N VPN Firewall and Cisco RV215W Wireless-N VPN Router could allow an unauthenticated, remote attacker to execute arbitrary code on an affected device. The vulnerability is due to improper validation of user-supplied input...
CVE-2020-3330		2020-07-16	Cisco Systems, Inc.	A vulnerability in the Telnet service of Cisco Small Business RV110W Wireless-N VPN Firewall Routers could allow an unauthenticated, remote attacker to take full control of the device with a high-privileged account. The vulnerability exists because a system account has a default and static password...
CVE-2020-3144		2020-07-16	Cisco Systems, Inc.	A vulnerability in the web-based management interface of the Cisco RV110W Wireless-N VPN Firewall, RV130W VPN Router, RV130W Wireless-N Multifunction VPN Router, and RV215W Wireless-N VPN Router could allow an unauthenticated, remote attacker to bypass authentication and execute arbitrary commands wi...
CVE-2019-...		2019-05-07	MITRE	An issue was discovered on Shenzhen Coship WM3300 WiFi Router 5.0.0.55 devices. The password reset functionality of the Wireless SSID doesn't require any type of authentication. By making a POST request to the

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I chose the first vulnerability that appeared:

CVE-2025-6237 - Path traversal & arbitrary file deletion in InvokeAI

CVE ID: CVE-2025-6237

Published: 2025-09-18.

Severity: **CRITICAL** - CVSS (CNA) **9.8 / 9.3** (different systems show 9.3–9.8).

Summary:

An unauthenticated path-traversal vulnerability in the InvokeAI bulk downloads API lets an attacker read and **delete arbitrary files** on the server by supplying crafted values to the `bulk_download_item_name` parameter.

Affected software version(s):

invoke-ai / InvokeAI: versions < **6.7.0** are affected. The issue is fixed in **6.7.0**.

Detailed description:

The vulnerable endpoint is: `GET /api/v1/images/download/{bulk_download_item_name}`. The server concatenates the user-provided `bulk_download_item_name` into a filesystem path **without canonicalizing or validating it**. By including traversal sequences (e.g. `../`) or absolute paths the attacker can escape the intended media directory and access or remove arbitrary files (SSH keys, DB files, configs, etc.). In some back-end/storage configurations this also enables overwriting or deletion of files referenced by the crafted path. CWE mapping: CWE-22 (Path Traversal) / CWE-73 (External Control of File Name or Path).

Impact & exploitability:

- **Impact:** Confidentiality, Integrity and Availability - attacker can read secrets (e.g., SSH keys), delete databases or configs, or otherwise disrupt the service.
- **Exploitability:** Network-accessible, requires **no authentication**, and has **low complexity** - trivial path-traversal strings are sufficient per advisory descriptions. That's why it's rated critical.

Direct references:

- NVD: CVE-2025-6237.
- GitHub Advisory (GHSA-vv9c-xxg7-wmv7) - includes link to the repo file and release.
- Vulnerable source file (example location in v6.0.0a1): `invokeai/app/api/routers/images.py` - lines referenced by advisories. Inspecting that file shows the code path that builds the download path.
- Huntr disclosure (original bounty/report): [huntr.com bounty page](https://huntr.com/bounties/2025-09-18).
- <https://cwe.mitre.org/data/definitions/73.html>

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2.11 Using results from the previous point (e.g. openssh 7.7 is vulnerable) search for the systems with this vulnerability (from point X) using

- A. <https://censys.io>
- B. <https://www.shodan.io>
- C. <https://zoomeye.org>

Write how many vulnerable systems have been found. Which countries are the 'top most' vulnerable?

Now we must check using Shodan, find systems exposing the InvokeAI endpoint that is vulnerable to CVE-2025-6237 and report counts and top countries.

We start by creating an account in shodan and > shodan init <api key>

I executed the following searches on Kali/terminal and via Shodan:

1. Check for the exact vulnerable endpoint: shodan count 'http.html:"/api/v1/images/download"'

```
(stud@kali-vm)-[~]
$ shodan count 'http.html:"/api/v1/images/download"'
0
```

2. Search for general InvokeAI fingerprint (title / body):

shodan count 'http.title:"InvokeAI"' / shodan count 'http.html:"InvokeAI"'

```
(stud@kali-vm)-[~]
$ shodan count 'http.title:"InvokeAI"'
2

(stud@kali-vm)-[~]
$ shodan count 'http.html:"InvokeAI"'
2
```

3. Attempt to list host details (IP / country):

shodan search --fields ip_str,port,location.country_name,org 'http.title:"InvokeAI" OR http.html:"InvokeAI"' --limit 1000

```
(stud@kali-vm)-[~]
$ shodan search --fields ip_str,port,location.country_name,org 'http.title:"InvokeAI" OR http.html:"InvokeAI"' --limit 1000 > invokeai_shodan.csv
Error: Access denied (403 Forbidden)
```

The error is a Shodan plan restriction preventing host-level exports from the current account (I have a free one).

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Summary:

- shodan count 'http.html: "/api/v1/images/download"' → 0
- shodan count 'http.title:"InvokeAI"' → 2
- shodan count 'http.html:"InvokeAI"' → 2
- Attempting shodan search to list IPs/countries returned **403 Forbidden** (free-tier limitation), so host metadata could not be exported from the CLI.

2 potentially exposed InvokeAI instances were found by Shodan. These two hosts are *potentially* vulnerable. Confirmation requires authorized version checks or safe manual testing.

Country-level information could not be retrieved from the current Shodan CLI session because shodan search/export is blocked by the account (403 Forbidden).

Therefore I cannot reliably list the top countries from the CLI output.

Finally, Shodan's results indicate *potential* exposure only; confirming CVE-2025-6237 requires authorized testing (version check, safe endpoint probes) on systems you own or are permitted to test. Never attempt exploitation on systems you do not control.